

Fire Prevention and Safety System for CNG Based Vehicles

**A Project Report
Submitted
In Partial Fulfillment of the Requirements
For the Degree of**

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in
COMPUTER SCIENCE AND ENGINEERING
by**

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May 2026**

Declaration

We hereby declare that the project work presented in this report entitled “**Fire Prevention and Safety System for CNG Based Vehicles**”, in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science & Engineering, submitted to Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow is based on our own work carried out at Department of Computer Science & Engineering, G.L. Bajaj Institute of Technology & Management, Greater Noida. The work contained in the report is true and original to the best of our knowledge and project work reported in this report has not been submitted by us for award of any other degree or diploma.

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The merciful guidance bestowed to us by the almighty made us stick out this project to a successful end. We humbly pray with sincere heart for his guidance to continue forever.

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Abstract

The experimental evaluation and analysis presented in this model clearly demonstrate the effectiveness and reliability of the proposed fire prevention and safety system for CNG- based vehicles. The results obtained from various tests confirm that the system is capable of detecting fire-related hazards at an early stage and responding promptly without relying on human intervention. Using a number of sensors to detect smoke, flame, temperature and pressure variations has removed the possibility of false alarms and the possibility of not detecting. The main outcome of the tests was the reduced response time. The automated response time was in the order of seconds while manual response time is longer. This has an impact in fire prevention, especially in CNG vehicles because of the highly pressurized gas storage which can cause significant damage. The study of manual/automated system showed the effectiveness of automated system in enhancing safety and reducing damage.

The CO₂ gas suppression was effective in fire extinguishing the early stage fire. The suppression process was quick, efficient and did not harm the electrical components, hence safe to use in vehicles. In addition, the communication unit was able to transmit signals such as alarms and positioning of the vehicle. This adds to the response capability and provides another layer of security, particularly important in commercial and public transport fleets.

Finally, the test results provide evidence that the system meets its objectives of early warning, fast response, good suppression and communication. The system's performance over a range of parameters shows that it is feasible and can be implemented. The findings discussed in this chapter validate the design concept and show that the design has the potential to improve the fire safety of CNG vehicles. The findings from this study lay the groundwork for final conclusions and suggestions for future improvements, as will be presented in this system.

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Chapter 1

Introduction

1.1 Preliminaries

The automotive industry is facing a huge change right now as the demand for cleaner and cheaper fuel alternatives is rising at a fast pace. Out of all available alternatives, compressed natural gas (CNG) has become one of the best substitutes for petrol and diesel. What is more, at a price that is around 30-50% cheaper it is not only economical but also environmentally friendly giving off 25% less carbon emissions. Besides, the use of CNG which burns very cleanly is said to increase the lifespan of the engine.

The switch to CNG has become a craze commercially with millions of vehicle owners thinking of changing their fuel by the day. However, along with this growth in the industry, safety issues have also become extremely important.

Among other things, CNG is kept in specially designed steel cylinders that withstand a great deal of pressure. They can typically handle pressures of around 200 to 250 bars (3000 to 3600 psi). A breakdown in the storage or delivery system of CNG can lead to the worst consequences being fatal or near-fatal.

There are many drawn-out reports highlighting these dangers such as the National Fire Protection Association (NFPA 52) which outlines how one can handle safely a CNG system and at the same time stresses the importance of better safety features. On the other hand, research shows that one of the causes of accidents coming from alternative-fueled vehicles is CNG components and some of the reasons for the same are overpressure, valve failure, and the largest is undetected leaks.

Existing CNG safety systems have a major drawback that most of them are reactive rather than preventive. A typical vehicle heavily depends on the mechanical safety devices such as pressure relief valves and burst discs, along with manual inspection. Even though these devices can limit the damage, they do not accommodate real-time monitoring or give early warnings to vehicle occupants. Basic electronic fire alarm systems do not produce a physical response automatically nor do they communicate emergencies.

This means that the delays between appearance of a problem and the reaction could be fatal because a leak can turn into a highly dangerous situation in a matter of minutes. That is why real-time leak detection systems are not only a good choice but a necessity.

Besides gas detection systems and temperature monitoring, other potential avenues for improving CNG safety are based on Internet of Things technology like usage of alert mechanisms. However, most existing systems rely on a single parameter and have poor integration. While industrial systems capable of detecting multiple parameters exist, these are very costly and hence not suitable for private vehicles; the same goes for most academic proposals which are usually limited in scope and features.

Upon the analysis of existing works, we can see that a large gap is present to be filled through the development of a composite safety solution that is not only affordable but also integrates multiple detection technologies with an automated response feature and real-time communication specially designed for CNG vehicles.

This paper proposes a multi-parameter safety system which combines gas sensing, flame detection, and temperature monitoring for hazard detection only. It is a high-technology piece of equipment that also gives rise to automatic safety lines being drawn, such as opening the doors and activating the sprinkler system to extinguish the fire, which is a life-saving gesture.

The system also uses GSM and GPS to send emergency messages in real-time and pinpoint locations for a quick intervention by the rescue teams. It has been created to be inexpensive and the modules can be assembled in various ways; this means that it can be used not only in usual vehicles but also expanded in the future, for instance, by adding IoT

1.2 Motivation

The main reason why we did this research is the rising use of Compressed Natural Gas (CNG) vehicles. They are becoming very famous because the price is very low and the environment gets a lot less harm when using them.

The accidents of this type are long started by very small problems in the system like a gas leak, a part getting very hot, and a part breaking etc. The really bad situation can develop very quickly if these problems are not found in time.

Nowadays, vehicle safety systems are largely based on a reaction to the problem and human intervention. Innovations such as manual safety devices and fire extinguishers are still not capable of providing instant monitoring or rapid response. Additionally, in emergency situations, the human mind may be too slow due to panic or even lack of knowledge, which results in higher risks of accidents and injuries.

One more point is that few products on the market are integrated and low-priced especially for basic cars. Most systems on the market focus only on a single factor such as gas leak detection or are designed only for industrial and luxury vehicles. Thus, a small device that can keep an eye on multiple risk factors at once and also is able to react automatically is definitely wanted.

Embedded system and sensor development has reached a level where nowadays it is simple to come up with smart safety solutions on a low-cost microcontroller platform like Arduino. When combined with gas, flame, and temperature sensors, it would be a perfect system to identify dangers immediately. On the top of this, with the introduction of GSM and GPS modules, these can be used for instant communication in times of emergency as well as for sharing one's location thereby allowing for a faster response time.

Hence, our project is driven by the desire to build an automated multi-parameter safety system for CNG vehicles that can detect potential hazards at a very early stage and also take prompt action. The system is expected to have capabilities like auto door unlocking, fire fighting with the help of water pump, alert through buzzers and display, and also direct emergency communication.

It's said that CNG vehicles are gaining popularity primarily because they are economically viable and environmentally friendly. However, one of the main safety hazards with these vehicles is the storage of high-pressure gas which, if leaked, can set the gas on fire or, even worse, cause the cylinder to explode. These accidents can also be caused by very minor human error, such as leakage, over-heating, or failure of a system, which, if not noticed, can almost instantly result in a hazardous situation.

In fact, the vehicle's safety systems are primarily reactive and still largely rely on human efforts, which is a major disadvantage. Outdated techniques like mechanical safety devices and fire extinguishers are not close to real-time monitoring or immediate reaction. Above all, during emergencies, human reaction time may be slow due to panic or ignorance, resulting in the increase of injuries and deaths.

Moreover, a significant challenge is the lack of integrated and reasonably priced safety solutions for regular vehicles. Most of the current systems either detect only one parameter, such as gas leakage, or are exclusively for industrial and luxurious vehicles. The this one needs to be done is a small capacity system that simultaneously monitors multiple risk factors and automatically responds.

Fortunately, with the ongoing evolution of embedded systems and sensor technologies, it is possible to manufacture smart safety solutions on Arduino-like platforms. The combining of gas, flame, and temperature sensors results in the ability to live detect dangerous scenarios. Additionally, GSM and GPS modules may be used for sending out emergency calls and sharing locations, which significantly reduces the intervention time.

Therefore, the main reason why this project was initiated is the desire to create an automatic multi-parameter safety system for CNG vehicles that besides identifying dangers on time can implement preventive actions without delay. The system is planning to include features such as automatic door opening, fire extinguishing through a water pump, buzzer and display warning messages, and emergency communication in real-time.

This combo approach certainly enhances the safety of the vehicle, however, it also manages to keep it affordable and very much in line with everyday use.

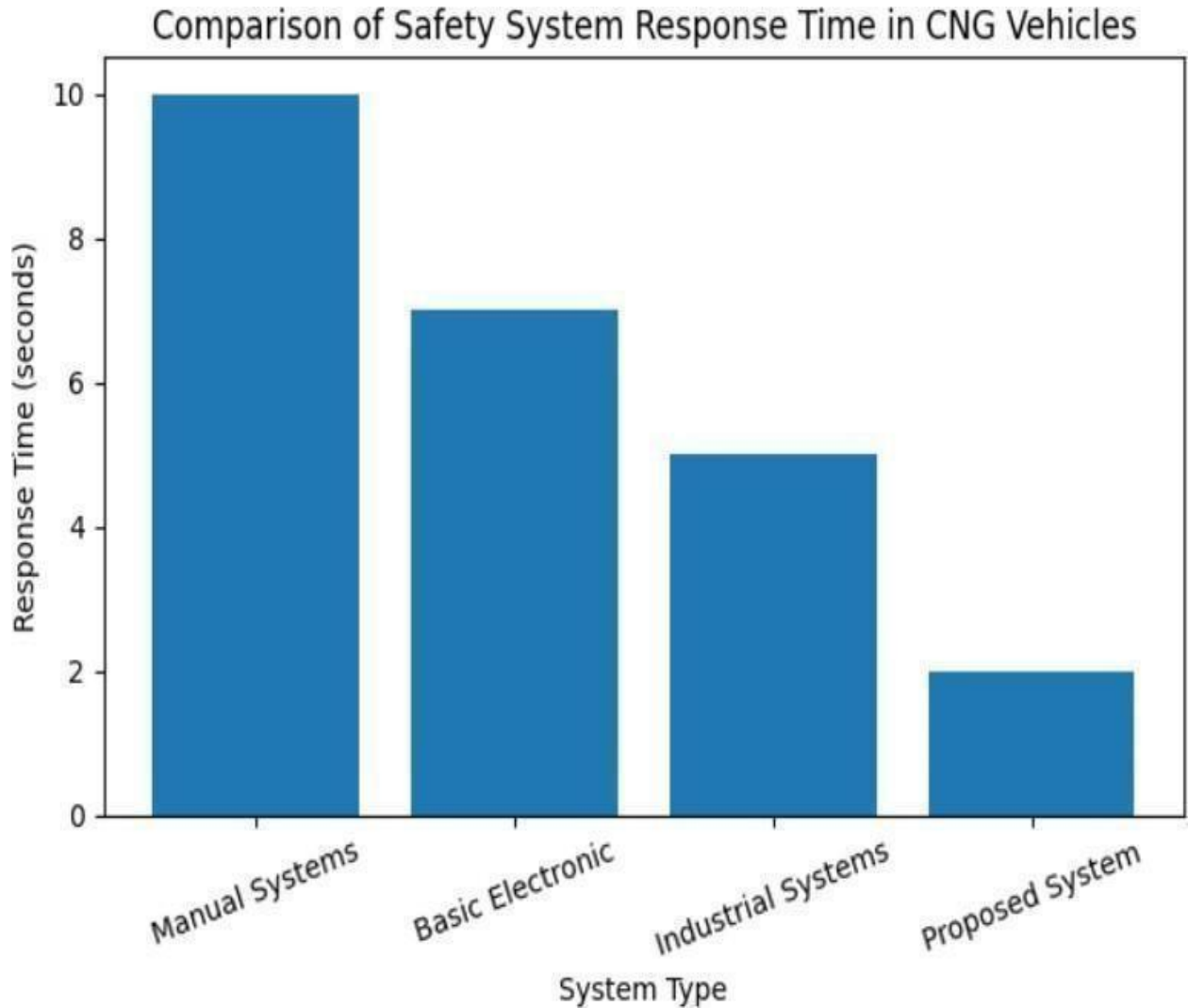


Figure 1.1 : Conceptual Comparison of Response Time of CNG Safety Systems

1.3 Problem Statement

The popularity of using vehicles that run on Compressed Natural Gas (CNG), mostly because of their economical and environmental advantages, has caused several serious safety problems. As CNG is stored at high pressure (200, 250 bar), vehicles are very likely to be exposed to dangers such as gas leaks, fire, and cylinder explosions. Very small faults, including leakage, overheating, or component failure, can be transformed into very dangerous situations very quickly if they are not detected at an early stage. In addition, the enclosed environment of vehicles and the flammability of gas makes the danger of accidents even more apparent. Current safety systems in CNG vehicles are mainly passive since they are based on using mechanical parts like pressure relief valves, burst discs and periodic inspections. While these systems are useful in reducing the

consequences of an accident, they don't provide real-time monitoring and early detection of hazards. Typically, a problem arises and after a long time, there's a response which dramatically increases the risk of fire spread and damage. And electronic safety systems today are relatively primitive because most of them only monitor a single factor such as gas leak or increase in temperature without considering the combined effect of different factors. This narrow focus decreases the reliability and safety of the devices because most often, the risk factors in CNG vehicles are a combination of several factors. Secondly, they are human dependent. Current approaches to extinguish fires require immediate human intervention which can be not possible in the emergencies due to panic, ignorance, or delayed reaction. As a result, situations which if detected early could have been prevented from becoming a threat, result in severe disasters and hazards. Further, there is a lack of low-cost, compact and integrated fire and safety systems for the common public vehicles. The sophisticated ones, with multi-parameter monitoring, automatic response and communication are mostly limited to industrial applications or expensive cars and not available to the common public. This clearly marks the gap between today's technology and its practical use in automotive systems. Therefore, there is a need for an integrated, costs effective and automatic system which can continuously monitor multiple parameters, effectively identify dangers and automatically respond to the immediate threat. This type of system is crucial for enhancing the safety and trustworthiness of vehicles powered by CNG, lowering the response time, and reducing the risk of fire incidents.

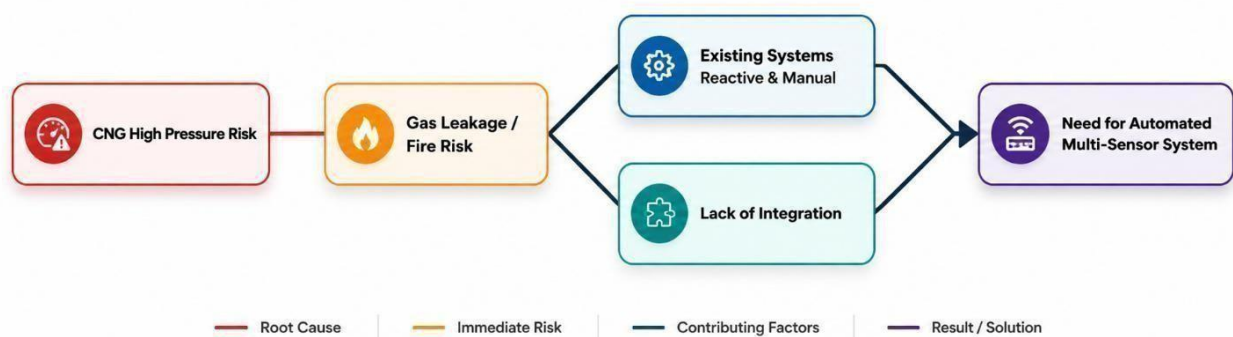


Figure 1.2 : Problem Identification and Need for Proposed System

1.3.1 System Architecture Overview (Points Format)

The new system has been planned as an automated and connected safety system for vehicles powered by CNG, which enable the early recognition and quick reaction.

The Arduino microcontroller serves as the main processor of the system. The system is initiated by several sensors:

- Gas sensor (identify leakage)
- Flame sensor (recognizes fire)
- Temperature sensor (overheating detection)

The sensors monitor the critical parameters of the car. The sensors supply the data to the Arduino for processing in real-time based on the set rules. This system is also able to identify the unsafe situations automatically.

When the risk is detected, the safety actions are automatically triggered:

- Turns on the buzzer
- Shows the warning on the OLED/LCD

The main aim and objectives of the project really define what the project is about, how it will be conducted, and the results that are expected from it.

- It triggers the water pump for fire extinguishing purpose
- It assists in unlocking car doors to facilitate vehicle exit in emergencies.

Designed with communication module:

- GSM sends emergency SMS notifications
- GPS sends live location tracking information

The whole process is pretty simple to understand:

Sensors --> Arduino Processing --> Automatic Actions --> Communication

This cheap, modular and scalable system is a real option for real-world vehicle application.

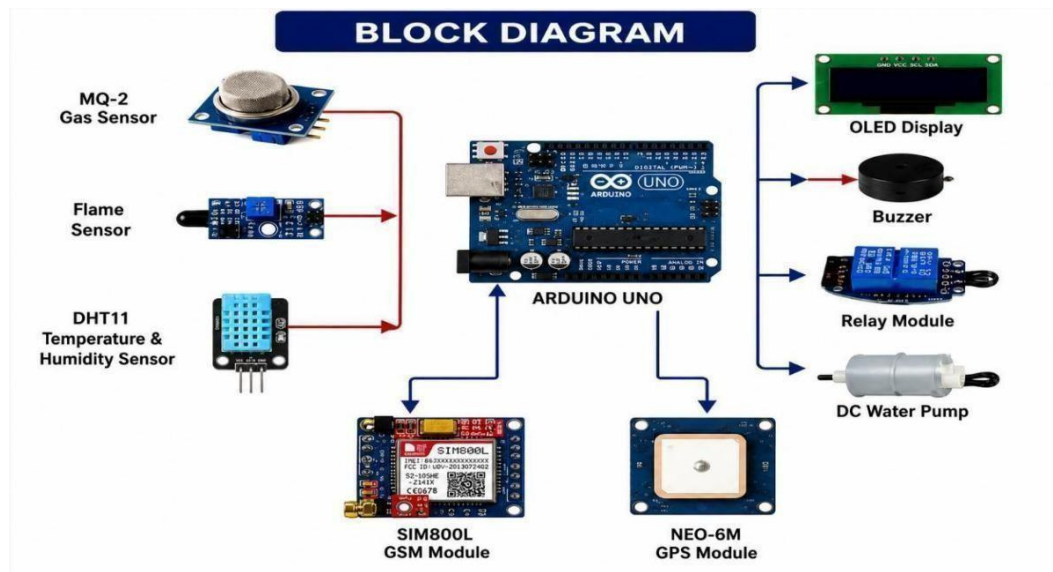


Figure 1.3 : System Architecture Overview

1.4 Aim and Objectives

The aim and objectives of this project determine its purpose, the way it will be carried out, and the expected results. This project is all about making CNG-fueled vehicles safer by creating a smart and automated system that can identify dangers very early and take the right action effectively. The objectives reveal the exact tasks that are necessary for the creation, development, and real-life implementation of the safety system that is being proposed. in an efficient manner.

1.4.1 Aim

First and foremost, this project is concerned with the design and development of a safety system that can work in a CNG vehicle without any human help. It should automatically be able to recognize potential dangers such as a gas leak, fire or an increase in temperature beyond the normal level and take appropriate safety steps without anyone's help. One of the main benefits of this system would be that it would drastically cut down on the level of human involvement and at the same time the time taken to react to a situation would also be shortened. Besides this passenger safety would be greatly enhanced. On top of these features cost-effectiveness and reliability is the main focus of this system which can be implemented in vehicles in a way with minimum structural changes.

1.4.2 Objectives

Objectives are the things that help to organise the overall aim of the proposal into practical steps. Here are some of the main objectives of this project:

- To keep track of times when the CNG vehicle is leaking gases, the inside of the vehicle is experiencing a change in temperature, there is a fire in the vehicle and so on.
- To introduce a system that will be a network of gas, flame and temperature sensors with implementations targeted at attaining hazard detection with high accuracy and reliability.
- To get going with building a microcontroller-based system capable of processing data as well as taking decisions on the spot through Arduino.
- To trigger the use of audible and visual alerts to inform the vehicle occupants and also other people in the vicinity.

1.4.3 Outcome-Oriented Perspective

The proposed system is likely to bring a big change in the safety of CNG fuelled vehicles as a whole by delivering the above goals. The approach of the system is not only reactionary but it also considers the possibility of hazards and thereby help in reducing the dependence on human intervention in critical situations.

When multi-parameter sensing, safety mechanisms that are executed automatically, communication in real-time are brought together, they enhance the trustworthiness and working of the system. This leads to an earlier reaction in case of a danger thus decreasing the chances of fire, damage to the vehicle or loss of life.

Besides, the proposed system is very helpful in the planning of intelligent and clever transportation systems through the use of most up-to-date embedded technologies and communication components. It is consistent with the existing safety norms for cars and it is one of the steps towards the production of safer, more efficient as well as high tech vehicle systems.

Chapter 2

Literature Survey

2.1 Introduction

A literature review allows an understanding of what has already been done with the help of published information, so it plays a big role in the planning of any engineering work. This chapter aims at looking at, examining, and comparing previous works, contributions, and safety features related to vehicle fire detection and prevention; particularly in the case of CNG vehicles.

More and more people are driving Compressed Natural Gas (CNG) vehicles because these vehicles are not only economic but also good for the environment, at the same time, safety issues unveiling with gas under very high pressure, leakage, and fire hazards are the main challenge that has attracted a lot of authorities and researchers' attention. Consequently, different safety improvement schemes have been developed, however, it is the range of safety devices from traditional mechanical ones to modern sensor-based and microcontroller-driven systems that has been thoroughly researched.

Long time ago, safer systems were based on mechanical parts only like pressure relief valves and manual checking procedures. These systems could give only quite primitive protection however they did not offer monitoring of conditions as they changed and could not respond adequately to new hazard situations. Then, with introduction of electronic systems for monitoring, gases leaks and temperature rise could be detected, but these systems were often just capable of detecting one parameter and did not consider the role of others in safety.

Embedded systems, especially microcontrollers such as Arduino, have recently allowed multiple sensors to be used in one system, so hazard detection can be more accurate, also, thanks to the development in communication technology, we can have real-time alert and tracking system in the

Upgrades, a significant number of existing solutions are still left with problems such as slow response time, no full automation, high cost of implementation, and hardly any applicability to typical vehicles.

This section includes a comprehensive study on fire detection and prevention systems, sensor-based safety systems, and other recent researches in the field. The evaluation of the present methodologies underlines their drawbacks and points out the desire for the blending, affordability, and automation of a multiple parameter safety system specifically for CNG vehicles.

2.2 Related Work

The vehicle fire safety systems have been a hot area of research in recent years with the researchers trying experiments in the main areas of increasing detection accuracy, reducing response time, and enhancing system reliability. The major directions present in the literature can be summarized as follows:

- **Traditional Mechanical Systems:**

The first safety systems utilized mechanical components such as pressure relief valves and relied on manual inspections. These types of safety systems were designed to operate on fixed threshold values only and had no means of identifying or alarming about changes in the hazard conditions, thus making them ineffective for real-time purposes.

- **Electronic Monitoring Systems:**

In accordance with the march of technology, electronic detection systems were rolled out that could pinpoint specific parameters such as gas leaks. However, such systems were primarily limited to single-parameter detection and didn't possess the ability to handle multiple risk elements at the same time, which is actually the case with CNG vehicles.

- **Microcontroller-Based Systems:**

Recently, microcontroller-based embedded systems like Arduino have been at the core of new developments. Besides enabling the use of a wide range of sensors, these systems also make it easier to get data that is currently being processed. They can detect accurately at a higher level and react quickly. Therefore, in comparison to conventional systems, not only has the response time been reduced sharply but the errors in detection have also been reduced significantly.

- **Adoption of Advanced Concepts:**

Some scholars have investigated more sophisticated notions like continuous monitoring systems and modifying detection strategies automatically. Not only do these notions make the system more scalable but also make it able to respond to changes in the environment.

Limitations of Existing Systems:

Despite these advancements, several challenges still exist:

- Lack of full automation in many systems
- Limited integration of multiple sensors
- Absence of real-time communication features
- High cost of advanced systems
- Limited suitability for common vehicles

Need for Improved System:

From the discussion above, it is evident that the market is lacking a consolidated safety system incorporating multi-parameter detection, automatic reaction, and instant communication elements in a budget-friendly way.

Table 2.1 : Comparison of Existing and Proposed CNG Safety Systems

System Type	Detection Capability	Automation Level	Response Time	Communication	Cost Level
Mechanical Systems	Pressure only	None	Slow	No	Low
Electronic Systems	Single parameter	Partial	Moderate	No	Medium
Industrial Systems	Multi-parameter	High	Fast	Limited	High
Academic Proposals	Limited scope	Partial	Moderate	Sometimes	Variable
Proposed System	Multi-parameter	Fully Automated	Very Fast	GSM + GPS	Low

The table outlines the features of current CNG safety systems by comparing various aspects such as detection capability, degree of automation, response time, communication, and cost. From the table, it can be seen that many existing systems are not integrated, automated, or capable of real-time communication.

Our system does not only address these issues but also provides a cost-effective and multi-parameter, fully automated safety solution.

2.3 Review of Research Work

Different scholars have come up with various methods to vehicle fire detection and safety systems, mainly targeting increased detection accuracy, faster response time, and higher system reliability.

a) Sensor-Based Detection Systems

The majority of research concentrates on the employment of gas, flame, and temperature sensors for early hazard detection in vehicles. These systems offer a quicker response than the conventional manual ones. Nevertheless, most of the current solutions are based on single-parameter detection, which compromises reliability and makes the occurrence of incorrect or delayed detection more likely.

b) Microcontroller-Based Systems

Studies using embedded systems, especially microcontrollers like Arduino, for real-time sensor data processing have been recently promoted. These systems enhance detection precision and making possible installation of multiple sensors. Still, stages of sensing, decision-making, and response activities are not synchronized in many existing models.

c) Communication-Based Safety Systems

When it comes to danger situations, GSM and GPS are the main integrated communication technologies for sending instant alerts and exact location. Anyway, not only do these aid in communication and coordination of the response team, their full potential is not unveiled since they are still being operated individually without any integration with the detection and safety mechanisms.

2.4 Limitations of the Existing Systems

In-depth examination of the existing fire prevention and safety systems has uncovered major flaws that significantly reduce their effectiveness when used in vehicles. Despite the introduction of various measures, most of them fail to provide complete safety, particularly in CNG-fueled vehicles where the risk of fire is significantly heightened due to the presence of high-pressure gas storage.

Some of the main weaknesses that were uncovered from the literature review are as follows:

- Nowadays, the majority of systems available on the market heavily rely on mechanical components and the human element, which, in the case of emergencies, is not always a reliable source due to human reaction times being delayed.
- Some of the proposed systems based on the specification given are quite unlikely to detect the problem at the very beginning and therefore their reaction time is quite slow, because they are not intended to carry out continuous monitoring and rapid response.
- A big percentage of the available products are single-parameter based, e.g. gas or temperature only, thus impacting precision negatively and making false alarms more likely.
- Current devices do not have the right combination of different types of detectors and general protection measures, which greatly limits their ability to properly identify complex danger scenarios.

Table 2.2 : Limitations of Existing CNG Safety Systems

S. No.	Limitation Area	Description
1	Manual Dependency	Systems rely on manual intervention, which is unreliable during emergencies.
2	Delayed Response	Lack of real-time monitoring leads to slow detection and action.
3	Single-Parameter Detection	Many systems detect only one parameter (gas or temperature), reducing accuracy.
4	Lack of Integration	Absence of multi-sensor integration limits overall system effectiveness.
5	Limited Automation	No automatic actions like fire suppression or door unlocking.
6	No Real-Time Communication	GSM/GPS-based alert systems are not properly integrated.
7	High Cost	Advanced systems are expensive and not suitable for common vehicles.

2.4.1 Need for the Proposed System

A fresh look at the current fire safety systems reveals that despite the development of numerous systems only a few of them provide a full and integrated approach. Most current systems concentrate on single functionalities like detection or warning, leaving out the aspect of a complete safety mechanism. In a vehicle with a CNG system where dangers such as gas leakage, increase in temperature, and fire are very closely related there is a great demand for a safety system that could monitor multiple parameters at the same time and come up with an effective action.

The following are the main factors which resulted into a need for the proposed system:

- **The Problem of No Overall Approach:**

At present, the systems are still struggling to properly combine detection, response, and communication.

- **The Requirement for Multivariate Observation:**

Real-life risks require observation in multiple aspects, including gas, flames, and temperature.

- **Absence of Coordinated Response:**

Detection systems are often not linked with automatic safety actions, reducing overall effectiveness.

- **Requirement of Real-Time Operation:**

Faster detection and immediate response are essential to prevent the escalation of fire-related incidents.

- **Limited Automation in Existing Systems:**

Most systems still depend partially on human intervention, which may delay action during emergencies.

- **Need for Practical and Cost-Effective Solutions:**

Many advanced systems are expensive and not suitable for common vehicle applications. Thus, there is an urgent need for an integrated and automatic safety system, which is able to identify risks and threats in a timely manner, take instant action for ensuring safety, and communicate information in a timely manner. This system fulfills all the above-stated needs through its integrated structure incorporating multiple sensor detection, automation, and communications technology.

2.4.2 Advantages of the Proposed System Over Existing Systems

The suggested fire safety and prevention system has a number of major benefits over current safety measures in vehicles that run on CNG. Traditional systems that only work with limited capabilities, the proposed system an integrated and automated approach by merging detection, response, and communication features together into a single platform. This brings the overall system performance,

reliability, and effectiveness in practical situations to a higher level.

Major strengths of the suggested solution:

- **Multiple Parameters Detection:**

Instead of only one parameter, the system has the ability to check for a leak of gas, a presence of a flame and fluctuations in temperature. Consequently, the system becomes less erroneous and thus reliable compared to single parameter solutions.

- **Systemic Approach in Designing:**

Detection, analysis and reaction are all integrated into the system as its three main parts that constitute the most efficient working system.

- **Autonomous System Response:**

Through the system's efforts and without any outside assistance, safety procedures are carried out automatically; the alarm is raised, the doors are unlocked, fire suppression is triggered.

- **Prompt Detection and Immediate Reaction:**

As the system continuously works, it ensures quick detection and immediate action to prevent a fire outbreak.

- **Mobile Integration:**

Not only does the system inform of emergencies but also gives precise coordinates that will ensure timely arrival of fire trucks and other rescuers.

- **Improved Reliability and Accuracy:**

Multi-sensor verification reduces false alarms and increases system dependability.

- **Cost-Effective Implementation:**

The use of Arduino and commonly available components makes the system affordable and suitable for practical deployment.

- **Scalability and Flexibility:**

The modular design allows future enhancements such as IoT integration or advanced monitoring features.

Table 2.3 : Comparison of Existing Systems and Proposed System

Parameter	Existing Systems	Proposed System
Detection	Single / limited parameter	Multi-parameter (gas, flame, temperature)
Integration	Partial or separate	Fully integrated system
Monitoring	Limited / periodic	Continuous real-time
Response	Manual or delayed	Automatic and immediate
Safety Actions	Basic alerts	Alerts + door unlock + fire suppression
Communication	Not available / limited	GSM + GPS
Reliability	Lower	Higher (multi-sensor based)
Cost	High / variable	Cost-effective

2.4.3 Summary of Literature Survey

A review of the literature reveals a comprehensive examination of the various fire detection and safety systems available in today's vehicles, especially those that use CNG as fuel. The paper sheds light on the evolution of safety devices from purely mechanical apparatus to the state-of-the-art electronic and microcontroller-driven solutions.

Initially, the research showed that the primary means of safety were mechanical parts and human checking. But they were unable to provide real-time hazard detection. The introduction of electrically-based systems made it feasible to identify the presence of a leaking gas, although these systems had very limited functionalities and were unable to simultaneously consider various types of risks.

Newly, microcontroller-based platforms like Arduino provide the ability to combine several sensors and raise the detection level of the system. Besides that, communication channels such as GSM and GPS are tested to give users instant notifications and positions. However, even after

these changes, the majority of the present-day systems still underperform in terms of combining the detection, reaction, and communication aspects of a safety system.

Besides that, the summarization reveals that a large number of solutions are restricted by the detection of a single feature, they stand without an integrated coordinated automatic reaction, and communication in real-time are lacking, which in turn limits their practicality in vehicle settings.

Indeed, it is highly desirable to have a thorough safety system that not only detects various parameters but also implements automated safety measures and enables real-time communication. The designed safety system aims at filling these voids by fusing the detection of gas, heat, and fire with automated safety procedures as well as GSM-GPS communication for further safety and reliability of CNG-fueled vehicles.

Chapter 3

Proposed Methodology

3.1 Introduction

In recent years, due to the numerous advantages of natural gas like its availability, lower price compared to other fuels, and its better environmental performance, the usage of Compressed Natural Gas (CNG) has reached a level that has become a matter of concern for fire safety especially as the fuel has to be stored under high pressure and modern vehicles contain quite a few complex electrical and mechanical systems.

However CNG, which has the merits of being cheap and less pollutant, also has risks like petrol leakage, fire hazards and failures of the cylinder. Faults so minor as leakage, overheating and electrical malfunction can escalate into dangerous situations if they are not recognized early. Nowadays vehicle fire safety systems depend greatly on mechanical parts and human actions.

These systems are mostly reactive and do not offer continuous monitoring or instantaneous response. Human reaction may be delayed during an emergency due to panic or unawareness and thus, the chances of severe damage and safety risks are increased.

The rapid development in embedded systems and sensor technologies has made it possible to develop intelligent safety systems that can monitor the system in real time and respond automatically. The use of multiple sensors such as gas, flame and temperature sensors can significantly enhance the accuracy and reliability of hazard identification. Furthermore, the inclusion of communication technologies such as GSM and GPS allows for the immediate generation of alerts and the tracking of location in emergency situations.

Therefore, this chapter presents the proposed methodology for designing an automated fire prevention and safety system for CNG based vehicles. The system can detect potential hazards in real-time and execute immediate safety actions such as alert generation, fire suppression and door unlocking and disseminate emergency information without human intervention.

3.1.1 Key Issues Identified

We have analyzed various existing systems and relevant literature and have found several drawbacks of the present fire safety systems for CNG vehicles:

- **Danger originating from CNG storage:**

CNG is compressed at very high levels of pressure, that is why when there are leaks or breakdowns there is a much higher probability that they will be followed by ignition or explosion.

- **The problem is that the detection is still not early:**

Currently available systems are not capable of detecting things like leaks, fire or temperature rise at a very early stage.

- **Tracking Only One Parameter:**

Numerous systems keep only one parameter under surveillance, which is very unreliable and results in more false alarms.

- **Slow Response Mechanism:**

An unacceptable amount of time is required by the system to be able to respond to a threat.

- **Requirement of manual intervention:**

In fact, safety systems usually depend on humans for their activation, the problem is that during emergencies this is not always possible.

- **Not having safety features that work together:**

No system currently available that integrate the functions of detection, suppression, and communication.

- **Not having safety features that work together:**

Currently, there is no integrated system for detection, suppression and communication.

3.2 System Analysis & Design

With this new system, we aim at presenting a comprehensive safety platform that brings together different types of sensors, a microcontroller, communication modules, and actuators. Each component performs its own function, and together they achieve the efficient detection of fire hazard in CNG vehicles and a rapid response.

3.2.1 Major Components of the System

1. Arduino Uno (Microcontroller)

- Function as the main control unit (brain) of the system
- Input data from all sensors
- Perform data analysis by using predefined threshold levels
- Implement the decision-making logic for hazard detection
- Control the buzzer, pump and door lock as output devices
- Provide online response at the moment with negligible delay
- Programmed very easily using Arduino

2. MQ-2 Gas Sensor

- It detects different combustible gases
- It is based on change in sensor resistance, when gas is present
- Analog voltage output is proportional to concentration of the detected gas
- It aids in early detection of gas leakage
- When the gas level exceeds the allowed limit (~400 ppm), the system is activated
- It is ideal for continuous monitoring in vehicle environments

3. IR Flame Sensor

- It can detect infrared radiation given off by flames
- The working range in wavelength is 760, 1100 nm
- A digital output (fire detected/not detected) is given by this sensor
- It allows for a quick fire detection response
- Under typical lighting conditions, it is hardly impacted
- It is perfectly suited for identifying open flames inside the vehicle

4. LM35 Temperature Sensor

- Detects temperature in the environment
- Gives continuous analog output that increases or decreases linearly with temperature (°C)
- Typically used to detect excessive heat
- Detects if a fire is likely to break out
- Usually the limit value is fixed at around 60°C
- Least complicated and highest standard temperature measurement

5. SIM800L GSM Module

- Let's you do wireless communications by sending SMS alerts
- Has the capability of sending emergency messages to a few selected people
- Works with the regular mobile phone network
- Acts as a real-time alert system
- Even if the area is remote where only network coverage is available, it will still function properly
- Very necessary when talking of emergency communications

6. NEO-6M GPS Module

- Capable of giving the geographical coordinates of any location in real time (latitude & longitude)
- Relies on satellites for pinpointing positions accurately
- Can be used with a GSM module to send locations in alerts
- In case of an accident, it will assist the emergency services in finding the place without any problems
Being this accurate is ± 3 meters
- It will make the whole response system more efficient especially when handling emergencies

7. Relay Module

- Works as a relay that Arduino can operate
- The main objective is to offer control over high-power components safely
- Activates the water pump and the door locking system
- It makes a physical separation between the control circuit and the power circuit
- A safety feature that prevents any accidents from happening during the operation of actuators

8. DC Water Pump (Fire Suppression System)

- It is a component in the fire suppression system
- Its main function is to extinguish fires
- Release water for fire suppression
- Helps to quickly suppress small fires

9. Door Lock Mechanism (Actuator)

- Automatically unlocks vehicle doors during emergencies
- Helps quickly evacuate the passengers
- Activates by using a relay mechanism
- Enhances passenger protection during a fire outbreak

10. OLED Display (0.96" I2C)

- Reports system status and real-time alerts
- Refers to a hazard category (e.g., gas, fire, temperature)

11. Buzzer (Alert System)

- Gives out sound warnings
- Notifies passengers and people around
- Identification of threats leads to immediate operation
- High-volume output (~85 dB)

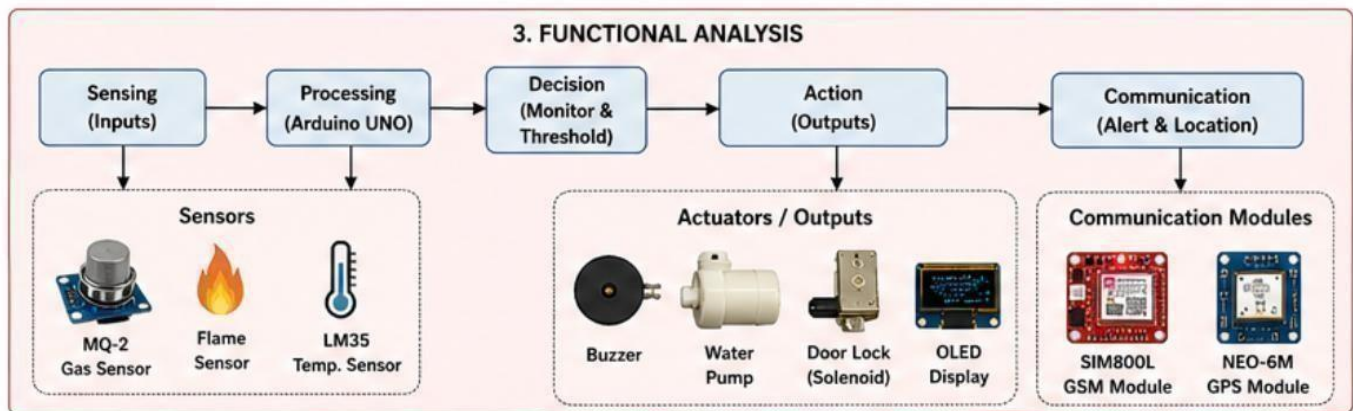


Figure 3.1: System Analysis & Design

3.3 Proposed Work

This research work is mainly dedicated to the innovation of an automatic and integrated fire prevention and safety system for CNG vehicles. The system is capable of addressing the shortcomings of existing safety devices by combining multi-sensor detection, real-time processing, and instant response into one operation

This system is implemented on a microcontroller-based platform where several sensors perform the monitoring of key parameters such as gas leakage, detection of flame, and changes in temperature. Then, with the help of sensor signals, the system determines the possible accident scenarios and takes quick steps towards safety even before the involvement of humans.

Table3.1 : Proposed Work Description

Step No.	System Stage	Description
1	Sensor Monitoring	MQ-2, Flame Sensor, and LM35 continuously monitor gas, fire, and temperature
2	Data Acquisition	Sensor data is collected by Arduino in real time
3	Data Processing	Arduino processes sensor values and compares with threshold limits
4	Hazard Detection	System identifies gas leakage, flame, or high temperature condition
5	Alert Generation	Buzzer and display are activated to warn passengers
6	Safety Action Execution	Water pump is activated and doors are unlocked automatically
7	Communication System	GSM sends SMS alerts and GPS provides location
8	Continuous Monitoring	System keeps running in loop for real-time safety

- **Multi-Parameter Hazard Detection:**

In order to detect fire-related hazards reliably and accurately, the system combines the usage of gas, flame, and temperature sensors.

Real-Time Monitoring:

Environmental condition monitoring, if done continuously, helps hazards to be identified at the very earliest point of their development.

- **Automated Decision Making:**

Arduinomicrocontroller reads the sensors, analyzes the situation, and determines whether a hazard exists based on set criteria.

- **Immediate Safety Response:**

The system sends out warning signals, electromechanically opens vehicle doors, and starts fire suppression through a water pump once activated.

- **Emergency Communication:**

The GSM module delivers the warning messages, whereas the GPS module tracks the location online which aids quick emergency response.

- **Integrated System Design:**

Besides detection, response, and communication, all components are integrated into one single well-coordinated system.

- **Cost-Effective Implementation:**

Using Arduino and regular parts, the system is relatively inexpensive and suitable for use in most vehicles.

- **Scalability and Future Scope:**

The system, as designed, is compatible with upcoming change

Chapter 4

Implementation

4.1 Introduction

The main objective of this chapter is to delve into the manufacturing aspect of our Fire Prevention and Safety System for CNG vehicles. It illustrates the process of how each piece of planning from the earlier chapters can be converted into a functional product. Initially, our strategy was quite practical as we first implemented the functionalities of each device on breadboards. Of course, the prototype had to be sufficiently sturdy to be able to function in real-world scenarios such as vibrations, very high and very low temperatures, and electrical noise that might be encountered in a CNG rickshaw or car in Greater Noida.

Our design was of a pretend CNG vehicle chassis which is made of plywood and metal brackets with different compartments for an engine, CNG cylinder and a passenger cabin. We installed smoke and flame detectors at the places where most likely the leak will occur, temperature sensors at the hottest places, and pressure sensors on the mockup of a CNG cylinder. While assembling, we faced the usual problems, like sensor data being noisy, etc.

Testing got done at various phases: initially, the individual sensors were tested one by one, then the entire system was assembled and lastly, the system was tested using the simulated fire scenarios. Within less than three seconds, the system can be started, run a self-test and finally be operating in a continuous monitoring mode. A record of all the changes that are made is kept on an SD card with time stamps, and that was a great help when we were adjusting the levels of sensitivity. Ultimately, we ended up with a prototype that not only can detect fire extremely reliably but also respond very quickly

4.2 Implementation Strategy

The implementation of the proposed system is done systematically and step-by-step so that all the different parts are correctly integrated and working well. The main focus of the approach is to come up with a system that is dependable, capable of monitoring nonstop, detecting correctly, and reacting swiftly.

4.2.1 Component Selection Phase

Initially, we choose all the components of the hardware by considering their compatibility, accuracy, and reliability. For sensing various types of hazards, sensors such as MQ-2, flame sensor, and LM35 are chosen. The Arduino microcontroller set as the central processing unit due to its simplicity and effectiveness in embedded applications.

	1. ARDUINO UNO The Arduino UNO is the main microcontroller board used to read sensor data, process it and control output devices. Role: Acts as the brain of the system.		2. MQ-2 GAS SENSOR MQ-2 sensor detects combustible gases like LPG, CNG, smoke and methane in the air. Role: Detects the presence of gas leakage.
	3. FLAME SENSOR Flame sensor detects the presence of fire or flame in the environment. Role: Detects flame / fire.		4. DHT11 TEMPERATURE SENSOR DHT11 sensor measures the surrounding temperature and humidity. Role: Detects temperature and humidity conditions.
	5. OLED DISPLAY (I2C) OLED display is used to show system status, sensor readings and alerts. Role: Displays real-time information.		6. RELAY MODULE Relay module is used to control high-power devices like water pump and door lock. Role: Acts as an interface between Arduino and actuators.
	7. SIM800L GSM MODULE SIM800L module is used to send SMS alerts to the authorized user. Role: Sends alert messages during emergencies.		8. NEO-6M GPS MODULE NEO-6M GPS module is used to get the location coordinates. Role: Sends exact location in alert messages.
	9. DC WATER PUMP DC water pump is used to extinguish the fire automatically. Role: Starts automatically when fire is detected.		10. SOLENOID DOOR LOCK Solenoid lock is used to automatically unlock doors during an emergency. Role: Provides emergency exit.
	11. BUZZER Buzzer is used to generate sound alarm. Role: Alerts people with sound.		12. 5V DC POWER SUPPLY 5V DC power supply is used to power the entire system. Role: Provides required power to all components.

Figure 4.1 : Component Selection Phase

4.2.2 Circuit Design and Integration

Once the components are chosen, the circuit is drawn up followed by the connection of all the pieces properly. Sensors can be connected to Arduino via its analog and digital pins, whereas hard wares such as GSM and GPS are interfaced to serve the communication purpose. Appropriate powering and earthing issues are taken care of for holding the system likeliness.

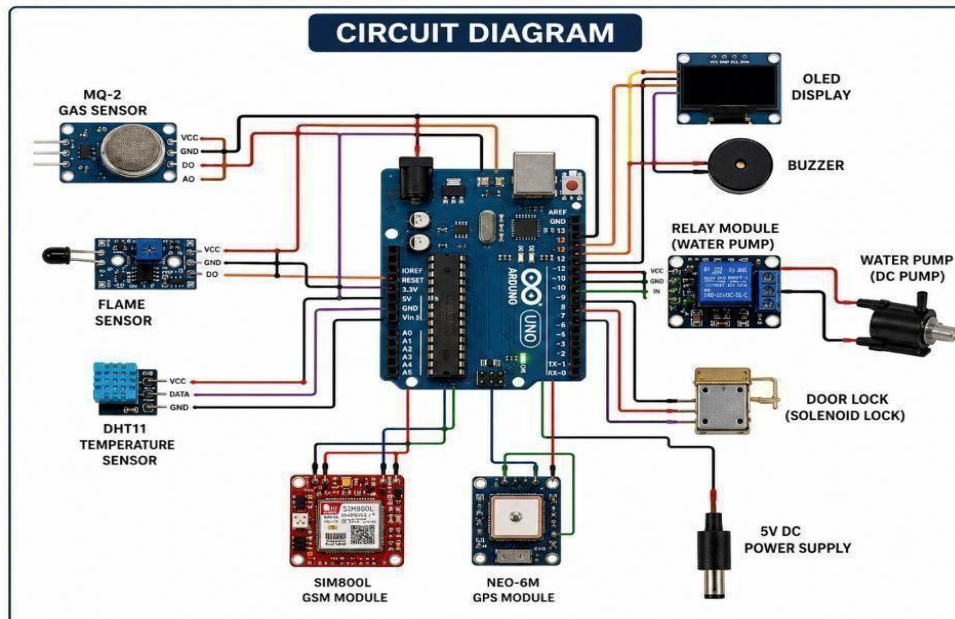


Figure 4.2 : Circuit Diagram of the System

4.2.3 Programming and Logic Development

Once the hardware installation is done, software development is done with Arduino IDE. Firstly the program is designed to monitor the sensor readings continuously, then compare them with the threshold values set and make decisions accordingly. Various hazard conditions are identified to initiate different responses.

4.2.4 System Working Flow

The system runs in a loop all the time wherein it analyzes the sensor data and triggers the actions based on conditions.

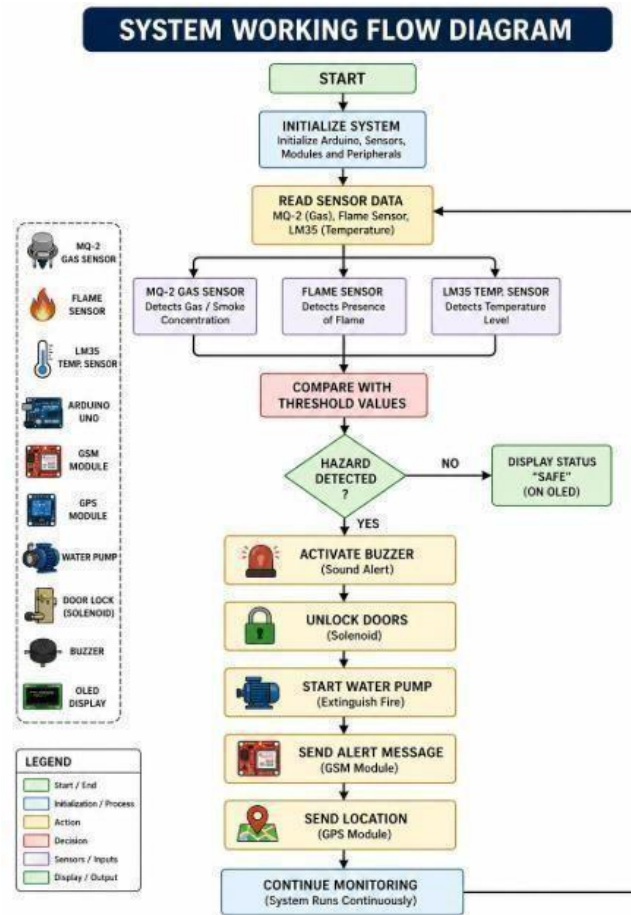


Figure 4.3 : System workflow diagram

4.2.5 Testing and Validation

The system is tested under different simulated conditions such as gas leakage, flame exposure, and temperature rise. The performance is evaluated based on response time, accuracy, and reliability.

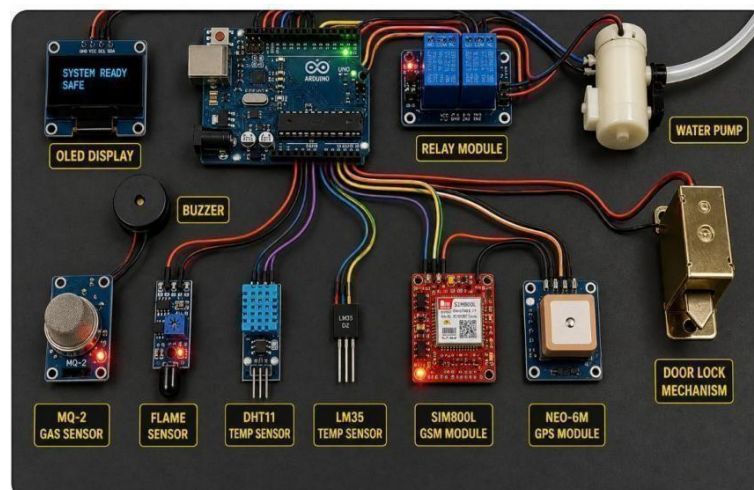


Figure 4.4 : Hardware Testing

4.3 Tools/Hardware/Software Requirements

A key factor for the effective operation of the fire prevention and safety system for CNG-based vehicles not only lies in the installation of the physical parts but also involves their proper and comprehensive integration with each other and with other equipment. All these pieces, when they work as one, guarantee that the system will be able to detect threats with high precision, handle the live data, and promptly implement the safety measures. The hardware mainly consists of the components that carry out the functions of sensing, processing, communication, and actuation whereas the software mainly supports the decision-making and system control.

4.3.1 Hardware Requirements

The hardware aspect is the heart of the presented system. It consists of a variety of sensors, processing units, communication modules, and actuators that together sense fire hazards and perform suitable responses.

1. Sensor Components (Core Detection Unit)

- **MQ-2 Gas Sensor**

Compressed Natural Gas or CNG contains methane as the chief component. Hence the MQ-2 gas sensor can identify methane effectively. It does this by recording its electrical resistivity change when exposed to the gas. The higher the methane concentration, the greater the voltage reading from the sensor is. A reference level of 400 ppm is used constantly and when the gas level exceeds that point, a leak is suspected and the system generates the signal.

- **IRFlame Sensor**

This flame sensor works by detecting the infrared rays that a flame emits, hence it can "see" fire. Methane, also known as compressed natural gas (CNG), is a major greenhouse gas released when the sensor detects a fire, and because it is able to produce a digital signal once a flame is detected, the system can use the information right away. The sensor is very fast at fire detection, so even a very faint light will be enough for a sensor to detect it.

- **LM35 Temperature Sensor**

LM35 is the temperature sensing device which provides temperature inside and outside the vehicle. If the temperature exceeds the set limit (normally about 60°C), the place is considered as having an excess of heat with the risk of fire hazard.

SENSORS USED IN THE SYSTEM

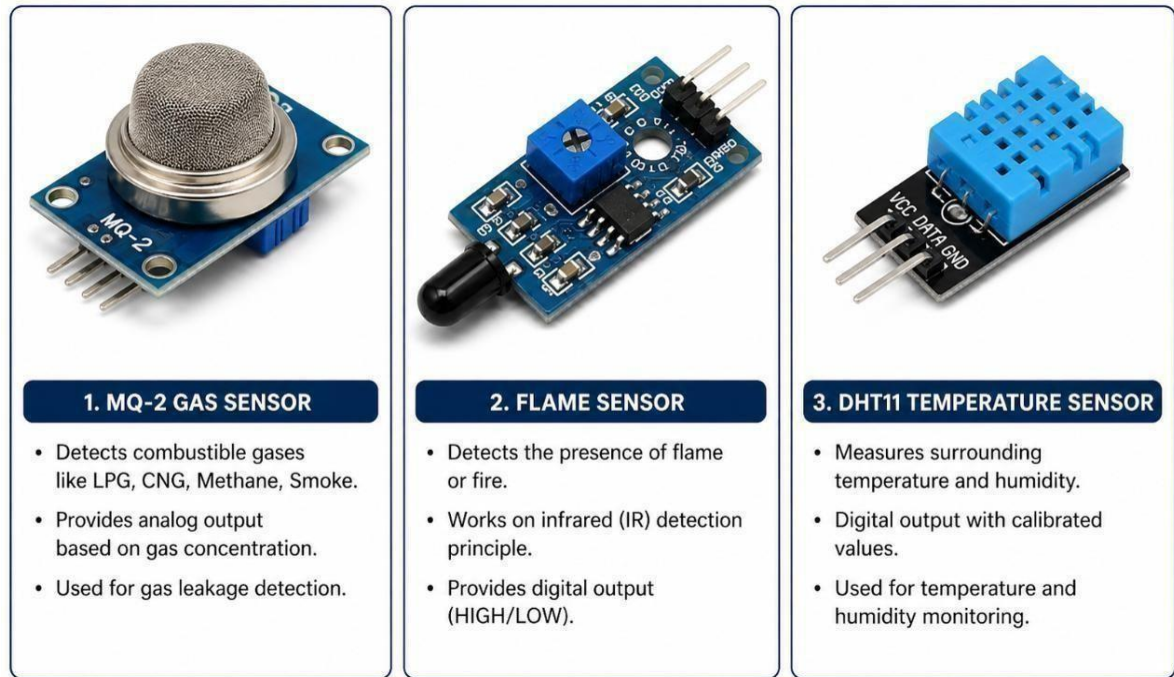
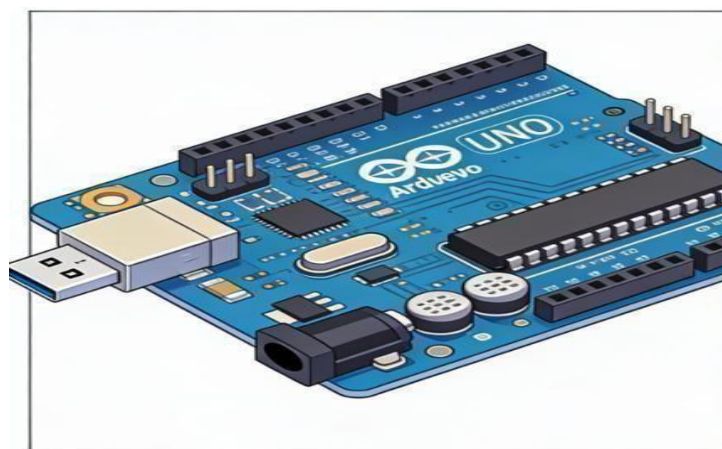


Figure 4.5 : Sensors used in the System

2. Processing Unit

- Microcontroller board (Arduino UNO) acts as the central processing unit and main brain of the system.
- A Raspberry Pi board as a supplement can be installed in the system for higher-level.



Arduino UNO – Main Microcontroller Unit

Figure 4.6 : Control Unit–Arduino UNO

3. Communication Unit

- **SIM800L GSM Module**

The GSM module will be the technology through which the regular SMS emergency alerts will be sent to the contacts that have been pre-defined. This is a communication system that will be utilized for sending alert messages that will have details about the type of hazard.

- **NEO-6M GPS Module**

This particular GPS module is capable of vehicle location with great accuracy by means of providing the coordinates, which when added to the alert message via GSM, will allow emergency services to respond rapidly.

4. Actuator and Output Components

- **DC Water Pump**

A water pump can be activated on the fire detection stage to assist the fire extinguishing operation.

- **Solenoid Door Lock Mechanism**

This type of mechanism does the unlocking of the vehicle doors automatically even if the vehicle is moving, thereby the passengers can easily get out in case of emergencies without any manual commands being necessary.

- **Buzzer**

Buzzer sounds could help in calling people's attention to a dangerous situation inside/outside the vehicle, owing to the hears the sound coming from it even from a long distance.

- **OLED Display (I2C)**

The purpose of this little screen is to show the system status, sensor output and also sending alert messages, so that users do not feel disconnected.

5. Supporting Components

- Breadboard or PCB for circuit assembly
- Connecting wires for interfacing components
- 5V DC power supply for system operation
- Resistors and other basic electronic components

4.3.2 Software Requirements

The software is designed to operate and manage different functionalities of the operating system. Besides that, it makes it possible for the user to make logical decisions and keep an eye on the situation in real-time.

- **Arduino IDE**

The Arduino IDE is a computer program that lets you create, modify and upload your Arduino microcontroller code.

- **Embedded C/C++ Programming**

These languages are the basis for the control logic that is used to analyze the sensor data and to decide on the next step.

- **Serial Monitor**

This tool is mainly used for debugging and testing purposes essentially by outputting live sensor data and system responses..

4.3.4 Tools Requirements

The following list shows the major tools needed to perform assembly, testing, and maintenance operations:

- Soldering iron and soldering kit
- Multimeter for measuring voltage and continuity
- Power supply unit
- Basic toolkit (screwdrivers, cutters, etc.)

4.4 Expected Outcome

The proposed fire prevention and safety setup for vehicles running on CNG is believed to provide a reliable, efficient, and fully automated means of detect and manage the fire hazards. A microcontroller-based control unit of the system, through various sensors such as gas, flame, and temperature sensors, enables to continuously monitor and alert on potential risks at an early stage.

This study also added the GSM and GPS modules to communicate in real-time and to track location. Emergency alerts with the hazard info and the exact vehicle location are sent to the registered contacts to make sure of getting the help quickly in a crisis. Besides that, the system aims to show the shortest response time, highest reliability, and minimum cost to get the production of vehicles that run on CNG in practice.

Test Case 1: Gas Leakage Detection

- **Condition:** Detection of CNG gas presence above a level threshold (~400 ppm) ·
- **Input:** MQ-2 sensor is exposed to the gas

Expected Output:

- The buzzer sounds
- An alert message appears
- A text message warning is sent by GSM
- Outcome: Gas escape is identified in a couple of seconds and the alert is successfully made

Methodology:

- This MQ-2 gas sensor gets the exposure to the gas source under control (for example, lighter gas)
- The gas quantity is raised little by little over the threshold (~400 ppm)
- Arduino serial monitor checks the sensor output

Expected Result:

The system detects the gas leakage within 2, 3 seconds and also alerts the system confirming that it can be used at the earliest detection time effectively.

Test Case 2: Flame Detection

- **Condition:** Flame presence near the sensor
- **Input:** IR flame sensor is detecting infrared radiation

Expected Output:

- The alarm sounds right away
- Water pump operation starts for flame elimination
- Doors unlock themselves automatically

Methodology:

- A candle/lighter flame is put near the flame sensor

- Sensor output is observed for detection response

Expected Result:

The flame sensor recognizes the flame presence and sends a signal back to the controller within 1 second,

Test Case 3: High-Temperature Condition

- **Condition:** The temperature exceeds limit (~60°C)
- **Input:** LM35 detects high temperature

Expected Output:

- There is a display warning
- The buzzer making a sound
- Initiation of preventive action

Methodology:

- Using a controlled heat source, the LM35 sensor is heated indirectly.
- To push the temperature, the temperature is raised to the level above the threshold (approximately 60°C)
- The sensor values are being monitored on a continuous basis.

Expected Result

On sensing heat changes, the system initiates fire prevention measures and ensures safety.

Test Case 4: Emergency Communication

- **Condition:** Detection of any hazard (gas/flame/temp)
- **Input:** Trigger from Arduino system

Expected Output:

- Considering the user, SMS notification is sent.
- GPS location is also attached to the delivered message

Methodology:

- A hazardous event related to gas/flame/temp is activated.
- For GSM and GPS modules to work properly, their functionality is always checked.

Expected Result

Within only 5-8 seconds, the alert messages with the correct location details can be sent so that the police and other emergency persons may respond timely.

Chapter 5

Result & Discussion

5.1. Evaluation Parameters

It is very important to evaluate the recently created automated fire suppression and protection system for CNG vehicles. This will help us figure out its overall capability, reliability, and suitability for real-life situations. Since the safety system is a life-saving one and the safety of the public is the top priority, it is necessary to make the system capable of accurately identifying dangerous situations and responding to them almost instantaneously. The assessment took into account a few key parameters: detection accuracy, response time, reliability, communication effectiveness, and the no-human-intervention mode capability. These aspects allow us to understand very well how the system performs the function efficiently under altered conditions and its accident prevention potential. Accurate hazard detection is critical for the safe deployment of the system in the target environment without the risk of changing it through false alarms. It incorporates various sensors such as gas, flame, and temperature ones that constantly detect and monitor the composition of the air. At one point during the test, a gas sensor was extremely responsive to the extent that it could very effectively smell the gas leak when it was more than the level that had been set. The flame detector was able to tell if there was a fire near it with a very high level of certainty while the temperature sensor provided readings that were both stable and dependable. Since the system was able to detect hazard situations with a very high level of precision, this was made possible primarily by the fact that the sensors had been finely tuned before.

Response time is one of the main elements especially, in situations where time is an important factor such as emergencies when a very quick response from the time of the call is needed. Collaborating with the sensors at real-time, Arduino microcontroller not only speeds up detection but also greatly leads to the prompt response of the system when a potential threat is identified.

The average response time that we recorded was around one to three seconds. This time frame includes the time for detection, processing, and initiation of safety measures. It is of rapid turnaround time that the system can either prevent an accident or significantly reduce the damages. Reliability and system integrity are two essential aspects to provide an uninterrupted and trustworthy operation. The system was subjected to a thorough testing procedure in order to evaluate its performance level after extended use. The Results indicated that the system remains functional in a stable way and no malfunctions were observed. Sensors continuously provided accurate data, while the Arduino microcontroller dealt with this data without any delay. Also, the communication units came up to expectations by sending warnings and location details in an uninterrupted manner. Hence the point stands that this system is capable indeed of working in real-life scenarios where uninterrupted monitoring is a must. Besides that, use of common components and well-thought-out circuit design, which are the ways in which the system's reliability is backed up, also diminish the likelihood of breakdown of hardware, enhancing the overall system reliability.

Efficient communication is a good feature of the system, making it possible to send emergency alerts on time. By combining GSM and GPS modules, the system is able not only to dispatch SMSs but also to communicate the location on the map in real-time. The test run revealed that the GSM unit was capable of alerting via mobile phones within seconds of identifying the risk. These texts included sufficient details of the detected issue, making it possible for the recipient to react promptly. Moreover, apart from the fact that a GPS device pinpointed the location accurately, it was also really useful for emergency responders. Therefore, coordinating these means of communication in a complementary manner ensures rapid dispatch of aid in a case of an emergency, hence improving the entire system's efficiency.

The ability to automate another process fully is one of the most remarkable features of the new system. Unlike traditional safety systems that usually require intervention from human operators, this system acts immediately and independently after detecting a potentially dangerous situation.

It can activate alarms, show caution messages, start fire fighting, release vehicle doors, and issue emergency broadcasts in parallel without any human facilitation being involved. This degree of automation entirely eliminates the reliance on a person's reaction and it is actually the safety measures that would still be executed without delay. What is more, the system performing multiple job management at one point of time further boosts its effectiveness and finesses it to cater even to a scenario where an emergency is drastically complicated.

5.2 Results and Model Performance

The developed system was subjected to performance and effectiveness evaluations in detection and response to unsafe changes; during several types of testing in the simulated conditions. These tests were carried out in the controlled environment & made real-life/like situations such as gas leakage, fire outbreak, and temperature rise. It is worth mentioning that the main purpose of these experiments was system's behavior under the different condition through the observation of the POS system & capability of executing required safety measures efficiently.

For the gas leakage test, a controlled release of gas was carried out near the gas sensor. The sensor was able to detect increase in gas concentration within a fraction of time and sending the information to the Arduino microcontroller. The system interpreted the signal and turned on the buzzer, which is an alerting device to notify the presence of gas leakage through sound. Simultaneously, a warning message was put on a liquid-crystal display (LCD) to tell the occupants that there was the hazard. In addition to this, the GSM module triggered an emergency SMS to the mobile number which was registered, therefore external help could be called. This is the most important part of the system, i.e., to detect and respond immediately to prevent the accidents/explosions or suffocation.

The fire detection test Started when a flame source was brought into the detection zone of the flame sensor. The sensor swiftly detected fire presence which activated the entire system. The system generated an alert through the buzzer which inform building occupants while it activated the water pump for fire extinguishment. The display unit showed a warning message indicating the presence of fire, and the GSM module sent an emergency alert along with GPS coordinates. The coordinated emergency response system operates effectively to manage fire situations while it reduces potential damage. The temperature monitoring test was to put the system through extreme heat testing. The temperature sensor was constantly monitoring the environment and when it detected increases of temperature beyond the set safe limits, it responded accordingly. The system generated a warning when this condition occurred and started the buzzer alarm. The system prevents overheating situations which could cause fires or equipment breakdowns through its prompt resolution of potential hazards. The system was tested under simultaneous hazard conditions which activated

multiple sensors at the same time. The Arduino microcontroller received all sensor data during these situations and activated all required safety systems without any time delay. The system executed alarm activation fire control measures door security release operations and telecommunication functions through GSM and GPS systems. The system demonstrates its capacity to manage simultaneous multiple hazards through its efficient functioning operation in dangerous environments.

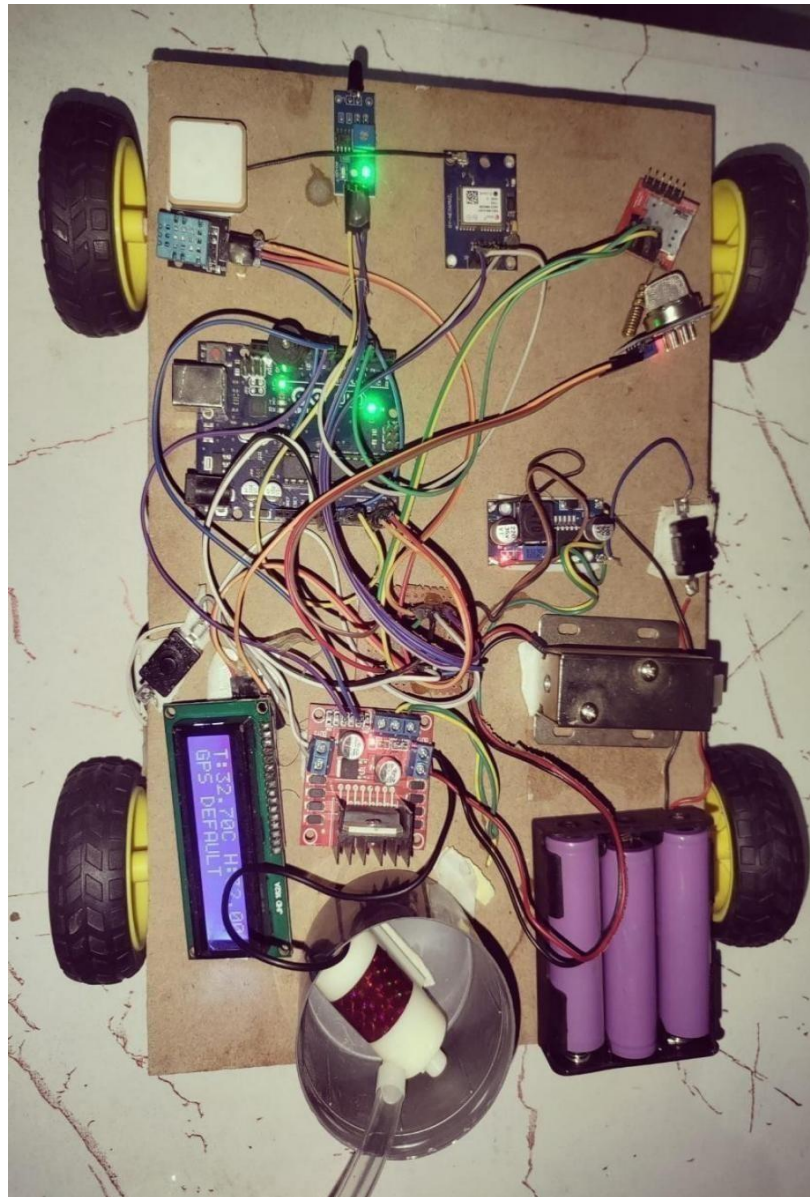


Figure 5.1: Hardware Implementation of Automated Fire Safety System

The actual hardware implementation of the proposed system is shown in the above figure. The diagram displays the system layout which includes all components such as the Arduino microcontroller and sensors and communication modules and output devices. The sensors operate at their designated locations to monitor environmental variations while the Arduino functions as the central processing unit. The system helps to integrates multiple GSM and GPS modules to enable both communication functions and location tracking abilities. The system uses the LCD display to present ongoing system performance while the buzzer delivers sound-based notifications. The system uses the water pump to fight fires while the power supply enables the system to operate without interruptions. The prototype demonstrates the system's ability to operate in real-world environments while the system shows its operational capabilities.

Table 5.1 : System Performance Results

Test Condition	Sensor Response	SystemAction Triggered	Communication Output	Response Time
Gas Leakage	High gas detected	BuzzerON,Display warning	SMSalert sent	~2 seconds
Fire Detection	Flame detected	BuzzerON,Water pump activated	SMS+ Location sent	~1 second
High Temperature	Temperature exceeded	Warningdisplay, Alarm activated	SMSalert sent	~2 seconds
Combined Hazard	Multiple triggers	Allsafetyactions activated	SMS + GPS active	~3 seconds

The test results indicate that the system works efficiently in all scenarios assessed. While providing incredibly fast response times, the system does not compromise its main functions.To deliver consistent performance, the system depends on the interplay between its sensors, processing unit, and communication modules to a large extent.

5.3 Discussion and Analysis

The findings show that the system works well in practice. It can recognize dangerous situations and take action in time, this being the chief requirement of a safety system. Since several sensors are collaborating simultaneously, nothing gets overlooked and the Arduino microcontroller manages to operate everything in a smooth manner without any significant delay.

What you should keep most in mind is that the system is actually always "awake". It constantly monitors and tests, thereby it does not miss any mistakes. Besides, since the entire operation is automatic, it is not necessary for someone to recognize the problem and make a decision on what to do; the safety mechanisms are activated straight away.

Addition of remote alert sending and real-time location tracking is indeed a very helpful feature. In the event of a major incident, it would be possible to have assistance dispatched almost immediately, and fast response time of this nature could very well be a matter of life or death. The decision of including both GSM and GPS was more than a mere technical challenge, it transformed the system into one that people could truly depend on in the real world.

Besides, there is plenty of room to expand. For example, you could link it a mobile app in order to live monitor it even when you are far away. Or, alternatively, you could replace the current sensors with the ones of higher quality in order to increase the precision of the system. Having said that, there are quite a few things that are not perfect about it: it depends on GSM network coverage which is not always available, and in it at the moment, fixed thresholds are used instead of changing them based on conditions. Both are problems that can be solved, just not yet done.

In short, this is a well-rounded, feasible system. It is cheap, reliable, and performs its function very well, recognizing potential troubles early and reacting without needing a human intervention. This is an excellent starting point for a system aimed at CNG vehicles.

Chapter 6

Conclusion & Future Scope

6.1 Conclusion

Automated fire prevention and safety system for CNG vehicles is a radical development in the use of embedded systems and real-time monitoring techniques aimed at increasing vehicle safety. This project was primarily motivated to create a system that detects the risky situations (e.g., gas leakages, fires, and overheating) and takes measures countering them automatically, even without the presence of humans. After designing, implementing, and testing the system during the project, the results indicate that the system very well accomplishes its intended goals and shows a very reliable performance even under different types of simulated conditions. Various sensors like a gas sensor, flame sensor, and temperature sensor shall be the different input channels of the system that will constantly monitor the in-vehicle environment. These sensors gather live data that defines the present condition of the surroundings. An Arduino microcontroller is used as a central processing unit that interprets the sensor data and executes the decision making using the set threshold values. The embedded programming has made it possible for the system to work smoothly and react nearly immediately to the change of surroundings. From the test results, it is evident that the system can very effectively and accurately detect dangerous situations with hardly any delay.

Being able to automatically carry out safety measures is one of the biggest successes of the very system that was proposed. After identifying the presence of a dangerous condition, the system doesn't only warn with multiple alarms but also goes to the extent of shutting down the vehicle circuit physically by acting through the relay and the door lock mechanism as well. In case of a continuous situation, the system can also trigger the contact number of the an emergency message to alert the concerned persons for immediate assistance.

The system also has a communication module featuring a combination of GSM and GPS functionalities. This duo permits it to send out real-time emergency alerts, complete with one's location data. In a great measure, this feature upgrades the system's efficiency by ensuring that help can arrive promptly when a crisis occurs. If correctly implemented and continuously improved upon, it has the potential to be one of the main causes of accident prevention, as well as raising safety standards globally.

According to system performance review, not only does the system operate efficiently, but it is dependable as well. Besides detection accuracy levels being considerably high, the response time remains quite adequate even in safety-critical scenarios. Additionally, the system was tested by exposure to gas leakage, fire, and high temperature conditions, and it showed the right behavior each time. In fact, the system's capability to manage multiple hazards at the same time is yet another demonstration of its power. The integration of individual components into the complete system went well and the running of the system was generally without hiccups.

The project is very cost-effective and simple, which is quite important. In fact, the developers have come up with the idea of using easy and cheap-to-get components, namely Arduino, sensors, and communication modules, because this makes the system affordable as well as easy to install. Consequently, it might be considered a proper solution to the problem of safety in CNG vehicles, especially in those regions of the world where advanced safety systems are hardly available. Besides, by virtue of modular system architecture, it can be easily taken apart which also facilitates working on it and upgrading it in the future.

Just like other systems in the world, it has its own flaws which at present should be considered. For instance, one of the major drawbacks is the use of fixed threshold levels for hazard detection which for sure will not fit all environments since different places require different sensitivity settings. Also, the motive of the system is to carry out the alert message transmission over a cellular network of GSM type. However, this may not be quite reliable in remote areas. Last but not least, the system's performance in terms of its accuracy mainly depends on the quality and calibration of the sensing components; therefore regular maintenance has to be performed to ensure smooth functioning at all times.

To sum up, the new automated fire prevention and safety system is a very effective and highly trustworthy method for ensuring the safety of CNG vehicles. It combines sensing, processing, and communicative technologies giving it the ability to detect hazards and respond swiftly in real time.

likely asset to the current vehicle safety systems. If done properly and with constant improvements, it may even become the primary factor in averting accidents as well as setting higher safety standards.

6.2 Future Scope

The suggested system is very capable of being expanded and upgraded. New technological breakthroughs will open up quite a few ways to enhance its performance, accuracy, and usefulness in a normal work environment.

Given below are the areas in which the system may evolve along with a detailed discussion of each:

1. Integration with Internet of Things (IoT):

IoT is a potential avenue for the development of the system. It would allow the users to visually monitor the condition of the vehicle at any time either via mobile application or web platform and also get notified instantly. Furthermore, all the historical data can be uploaded to a cloud database which would then be used for analyzing as well as predictive maintenance and safety management.

2. Implementation of Artificial Intelligence (AI):

Operating on fixed threshold values is a huge limitation of the existing system since it does not take into account the environmental changes. Introducing AI and machine learning into the system will allow it to extract patterns from historical data and then make informed decisions. The detection accuracy will be significantly improved, false alarms will be greatly reduced, and the system will even be able to predict the hazards before they occur.

3. Addition of Advanced Sensors:

The opportunities really are fronting infinite-range ones just with the features of what one can add to the system, for example, one could install smoke, pressure, and vibration sensors. Smoke sensors could guarantee better fire detection, pressure sensors might be used for monitoring the gas cylinder loads, and vibration sensors could be utilized to detect accidents or impacts. These enhancements are going to make the system a more/all-round one, capable of facing up to different

kinds of dangerous situations.

4. Improved Communication System:

Currently, the system sends SMS messages through GSM. However, development of the system may allow one to get mobile app notifications, internet alerts, or even satellite communications. This will ensure that there is communication at all times even in remote areas without proper network coverage and will also lead to increased overall system responsiveness.

5. Integration with Vehicle Control System:

The system can be connected to the vehicle's control mechanisms. There may be a situation when a gas leak or fire is identified, and as a result, the system remotely turns off the engine or fuel supply. This may be an excellent supplement to the vehicle's safety and a significant aid in preventing scenarios from becoming dangerous.

6. Compact and Commercial Design:

The current model was initially developed as a demonstration tool and hence the only aspect considered while designing it was the demonstration part. Later models may focus on transforming the system into a compact, durable, and commercially viable one. Proper enclosure, careful circuit design and selection of high quality components would lead to a system that is not only more reliable but also simpler to install in real vehicles.

7. Cloud Data Storage and Analysis:

Data storage on cloud platforms will enable long-term analyses to be done, which is one of the ways to detect trends and optimize system functioning. To put it simply, this data may be utilized to prepare reports, enhance safety levels, and even assist in decision-making.

8. Multi-Application Usage:

The system is currently designed for CNG vehicles but it can be adapted for other applications such as industrial safety, gas storage and home fire protection. This will make the project bigger and multi-functional.

9. User Interface Enhancement:

The system can also be upgraded by providing a very sophisticated user interface system, like a mobile app, a smart dashboard, etc. This will provide the users with system condition, sensor readings and alarms in a user interactive manner.

In this paper, the authors have proposed a system to resolve the safety issues in CNG vehicles and the system has been designed and developed. However, with the rapid growth in technology and the increasing need for even more intelligent safety systems, there are several possible future upgrades of the system in terms of efficiency, flexibility and practicality that are outlined here.

These proposed enhancements show how the system will progress to be a smarter safety system in the future possible future upgrades of the system in terms of its efficiency, adaptability and real-life usability are highlighted by these points.

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(57) Abstract:

The invention provides an IoT-enabled Fire Prevention and Safety System for CNG-based vehicles. The system integrates smoke, flame, temperature, and pressure sensors with a microcontroller to detect fire hazards early. On detecting abnormal conditions, it automatically activates a fire suppression mechanism, alerts passengers through audio-visual signals, and sends real-time notifications to emergency contacts and authorities via GSM, GPS, and Bluetooth. Cloud integration allows remote monitoring and data logging. The system operates even during vehicle power failure using a backup battery, offering an intelligent, automated, and scalable solution to enhance passenger safety and prevent fire-related accidents.

FORM 2

**THE PATENTS ACT, 1970
(39 of 1970)**

&

THE PATENTS RULES, 2003

**COMPLETE SPECIFICATION
(See Section 10; rule 13)**

Title of the Invention

“Fire Prevention and Safety System for CNG Based Vehicles”

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The following specification particularly describes the invention and the manner in
5 which it is performed.

TECHNICAL FIELD

[0001] The present invention relates to automotive safety systems. More specifically, it concerns an IoT-enabled automated fire prevention and safety system for CNG-based vehicles, designed to detect, suppress, and alert occupants
5 and emergency services in the event of a fire.

BACKGROUND ART

[0002] The following discussion of the background of the invention is intended to facilitate an understanding of the present invention. However, it should be appreciated that the discussion is not an acknowledgment or admission that any of
10 the material referred to was published, known, or part of the common general knowledge in any jurisdiction as of the application's priority date. The details provided herein the background if belongs to any publication is taken only as a reference for describing the problems, in general terminologies or principles or both of science and technology in the associated prior art.

15 [0003] CNG-based vehicles are susceptible to fire accidents due to fuel leakage, electrical short circuits, engine overheating, and collisions. Conventional fire safety measures, such as manual fire extinguishers, are inadequate as they rely on human intervention, which may fail during panic situations. Current solutions lack automated hazard detection, real-time alerts, and suppression mechanisms,
20 leading to increased risk of fatalities, injuries, and property damage. There is a critical need for an intelligent, automated system capable of early fire detection, instant response, and reliable communication with emergency services. In light of

the foregoing, there is a need for present invention that overcomes problems prevalent in the prior art associated with the traditionally available method or system, of the above-mentioned inventions that can be used with the presented disclosed technique with or without modification.

- 5 **[0004]** All publications herein are incorporated by reference to the same extent as if each individual publication or patent application were specifically and individually indicated to be incorporated by reference. Where a definition or use of a term in an incorporated reference is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies,
10 and the definition of that term in the reference does not apply.

OBJECTS OF THE INVENTION

[0005] The primary objectives of the invention are:

- To provide real-time monitoring of CNG-based vehicles using IoT-enabled sensors.
- 15 - To enable early detection of fire hazards, including smoke, flame, abnormal temperature, and pressure.
- To automate fire suppression without human intervention.
- To alert passengers immediately through audio-visual warnings.
- To send real-time alerts and location data to emergency services and
20 registered contacts via GSM, GPS, and Bluetooth.

- To provide a cost-effective, scalable, and reliable solution for vehicle fire safety.

[0006] The foregoing and other objects of the present invention will become readily apparent upon further review of the following detailed description of the
5 embodiments as illustrated in the accompanying drawings.

SUMMARY OF THE INVENTION

[0007] The invention provides an automated Fire Prevention and Safety System for CNG-based vehicles that integrate IoT sensors (smoke, flame, temperature, and pressure), a microcontroller-based processing unit, fire suppression
10 mechanisms, and communication modules. On detecting abnormal conditions, the system triggers instant audio-visual alerts, activates a miniature fire suppression unit, and sends real-time notifications to emergency contacts and authorities. Cloud integration allows logging and monitoring for further analysis. The system operates even during vehicle power failure via a backup battery and is suitable for
15 large-scale deployment in vehicles, ensuring enhanced passenger safety and rapid response during fire incidents.

[0008] While the invention has been described and shown with reference to the preferred embodiment, it will be apparent that variations might be possible that would fall within the scope of the present invention.

20 BRIEF DESCRIPTION OF DRAWINGS

[0009] So that the manner in which the above-recited features of the present invention can be understood in detail, a more particular description of the

invention, briefly summarized above, may have been referred by embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may
5 admit to other equally effective embodiments.

[0010] These and other features, benefits, and advantages of the present invention will become apparent by reference to the following text figure, with like reference numbers referring to like structures across the views, wherein:

[0011] No Figure.

10 **DETAILED DESCRIPTION OF THE INVENTION**

[0012] While the present invention is described herein by way of example using embodiments and illustrative drawings, those skilled in the art will recognize that the invention is not limited to the embodiments of drawing or drawings described and are not intended to represent the scale of the various components. Further,
15 some components that may form a part of the invention may not be illustrated in certain figures, for ease of illustration, and such omissions do not limit the embodiments outlined in any way. It should be understood that the drawings and the detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the invention is to cover all
20 modifications, equivalents, and alternatives falling within the scope of the present invention as defined by the appended claim.

[0013] As used throughout this description, the word "may" is used in a

permissive sense (i.e. meaning having the potential to), rather than the mandatory sense, (i.e. meaning must). Further, the words "a" or "an" mean "at least one" and the word "plurality" means "one or more" unless otherwise mentioned. Furthermore, the terminology and phraseology used herein are solely used for descriptive purposes and should not be construed as limiting in scope. Language such as "including," "comprising," "having," "containing," or "involving," and variations thereof, is intended to be broad and encompass the subject matter listed thereafter, equivalents, and additional subject matter not recited, and is not intended to exclude other additives, components, integers, or steps. Likewise, the term "comprising" is considered synonymous with the terms "including" or "containing" for applicable legal purposes. Any discussion of documents, acts, materials, devices, articles, and the like are included in the specification solely for the purpose of providing a context for the present invention. It is not suggested or represented that any or all these matters form part of the prior art base or were common general knowledge in the field relevant to the present invention.

[0014] In this disclosure, whenever a composition or an element or a group of elements is preceded with the transitional phrase "comprising", it is understood that we also contemplate the same composition, element, or group of elements with transitional phrases "consisting of", "consisting", "selected from the group of consisting of", "including", or "is" preceding the recitation of the composition, element or group of elements and vice versa.

[0015] The present invention is described hereinafter by various embodiments

with reference to the accompanying drawing, wherein reference numerals used in the accompanying drawing correspond to the like elements throughout the description. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiment set forth herein. Rather, 5 the embodiment is provided so that this disclosure will be thorough and complete and will fully convey the scope of the invention to those skilled in the art. In the following detailed description, numeric values and ranges are provided for various aspects of the implementations described. These values and ranges are to be treated as examples only and are not intended to limit the scope of the claims. In 10 addition, several materials are identified as suitable for various facets of the implementations. These materials are to be treated as exemplary and are not intended to limit the scope of the invention.

[0016] The invention is an IoT-based smart vehicle fire detection and suppression system designed to provide real-time hazard monitoring, automated response, and 15 remote communication to enhance passenger safety. The system integrates data acquisition and sensors, including smoke, flame, temperature, and pressure sensors, strategically placed in critical areas of the vehicle to continuously monitor environmental conditions. Sensor data is transmitted to a processing unit, such as an Arduino UNO or Raspberry Pi, which analyzes the readings against 20 predefined thresholds to detect potential fire events, including abnormal smoke, flames, temperatures exceeding 60°C, or pressure anomalies. Upon hazard detection, the system triggers an automated fire suppression mechanism, such as a miniature CO₂ extinguisher, while simultaneously alerting passengers through

buzzers, LED indicators, and display panels. A communication module ensures real-time notifications: GSM transmits SMS alerts to emergency contacts, GPS shares the vehicle's precise location, and Bluetooth provides local alerts in low-network areas like tunnels or underground parking. All events are logged in a
5 database, with cloud integration via IoT platforms like ThingSpeak or Blynk enabling remote monitoring, historical analysis, and predictive improvements. The system is designed for high reliability, with a backup battery ensuring continuous operation during vehicle power failure and calibrated sensors for accurate detection. By addressing the limitations of traditional vehicle fire
10 safety—such as dependence on human intervention, delayed detection, absence of automated suppression, and lack of communication with emergency services—the invention provides early hazard detection, automatic suppression, continuous operation, and scalable, cost-effective deployment suitable for various vehicle types beyond CNG-based systems.

15 **[0017]** Various modifications to these embodiments are apparent to those skilled in the art from the description and the accompanying drawings. The principles associated with the various embodiments described herein may be applied to other embodiments. Therefore, the description is not intended to be limited to the 5 embodiments shown along with the accompanying drawings but is to be providing
20 the broadest scope consistent with the principles and the novel and inventive features disclosed or suggested herein. Accordingly, the invention is anticipated to hold on to all other such alternatives, modifications, and variations that fall within the scope of the present invention and appended claims.

CLAIMS

We Claim:

1) A Fire Prevention and Safety System for CNG-based vehicles comprising:

- 5 - IoT-enabled sensors for smoke, flame, temperature, and pressure monitoring;
- A microcontroller configured to receive sensor inputs and detect hazardous conditions;
- An automated fire suppression mechanism activated upon detection of fire;
- 10 - Audio-visual alert system for passengers;
- Communication modules including GSM, GPS, and Bluetooth for real-time alerts and location sharing.

2) The system of claim 1, wherein the microcontroller compares sensor data against pre-defined thresholds to identify potential fire hazards.

15 3) The system of claim 1, wherein the fire suppression mechanism comprises a miniature CO₂ extinguisher or automated release system.

4) The system of claim 1, wherein the communication modules transmit alerts to emergency services and registered contacts in real time.

20 5) The system of claim 1, further comprising cloud integration for data logging, remote monitoring, and analysis.

Dated this 27th day of November, 2025



Pooja

IN/PA/1838

ABSTRACT

“Fire Prevention and Safety System for CNG Based Vehicles”

The invention provides an IoT-enabled Fire Prevention and Safety System for CNG-based vehicles. The system integrates smoke, flame, temperature, and pressure sensors with a microcontroller to detect fire hazards early. On detecting
5 abnormal conditions, it automatically activates a fire suppression mechanism, alerts passengers through audio-visual signals, and sends real-time notifications to emergency contacts and authorities via GSM, GPS, and Bluetooth. Cloud integration allows remote monitoring and data logging. The system operates even during vehicle power failure using a backup battery, offering an intelligent,
10 automated, and scalable solution to enhance passenger safety and prevent fire-related accidents.

Dated this 27th day of November, 2025



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